

# Sanskrit-to-English Neural Machine Translation

## Assignment 2 - Natural Language Understanding

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[github.com/G25AIT1012/Natural-Language-Understanding-Assignment-2](https://github.com/G25AIT1012/Natural-Language-Understanding-Assignment-2)

### 1. Introduction

1. This assignment builds a system that translates Sanskrit into English.
2. Sanskrit is hard to translate because it is highly inflected and has very little parallel data.
3. Training a model from scratch on only 10,000 pairs would not work well, so I fine-tuned a pretrained multilingual model instead.
4. I used Meta's NLLB-200 (distilled, 600M parameters), which already supports both Sanskrit and English.
5. Fine-tuning adapts it to this dataset.
6. On the test set it reached a corpus BLEU of 0.2615 and a BERTScore (F1) of 0.6343.
7. All pretrained parts are listed in Section 9, and no external APIs were used.

### 2. Architecture

NLLB-200 is an encoder-decoder Transformer. I did not change the architecture; I only fine-tuned it.

1. **Encoder:** reads the Sanskrit sentence using multi-head self-attention and feed-forward layers, and turns it into contextual vectors.
2. **Decoder:** generates the English translation one token at a time, using masked self-attention over what it has produced, cross-attention over the encoder output, and a feed-forward layer.
3. **Attention:** lets each word attend to the others, which helps with Sanskrit's word agreement and long-distance dependencies.
4. **Language and decoding:** I set the source language to san\_Deva and force the first output token to eng\_Latn so the model produces English. Decoding uses beam search (beam 6, length penalty 1.0, no-repeat n-gram 3).

### 3. Training Procedure

1. **Preprocessing:** aligned the Sanskrit and English files on Source\_id and removed empty rows.
2. **Tokenization:** used NLLB's own SentencePiece tokenizer (max length 128 tokens).
3. **Hyperparameters:** 10 epochs, learning rate 2e-5, batch size 8 with gradient accumulation 2 (effective 16), warmup ratio 0.10.
4. **Optimization:** HuggingFace Seq2SeqTrainer with AdamW and a warmup-then-decay schedule; fp16 and gradient checkpointing to fit the model on the GPU.
5. **Regularization:** label smoothing 0.10, weight decay 0.01, and early stopping (patience 2); the best checkpoint on dev BLEU was kept.

### 4. Results

1. Results are on the test set (1,000 sentences).
2. BLEU uses NLTK with default weights; BERTScore uses the official bert-score library (F1, rescaled with baseline).

| Metric        | Value (test set) |
|---------------|------------------|
| Corpus BLEU   | 0.2615           |
| Sentence BLEU | 0.2301           |

|                                  |               |
|----------------------------------|---------------|
| BERTScore (F1)                   | 0.6343        |
| Inference time (1,000 sentences) | 94.64 seconds |
| Model parameters                 | 615,073,792   |

3. Inference ran on a single NVIDIA A100 (40 GB) GPU, using about 16.8 GB of GPU memory.
4. The dev set was used during training to pick the best checkpoint.
5. Training and validation loss went down each epoch while dev BLEU went up, until early stopping; the full log is in the notebook.
6. BLEU (~0.26) is much lower than BERTScore (~0.63).
7. This is because the model often gives a correct translation using different words than the reference, which BLEU punishes but BERTScore does not.
8. The examples below explain this.

#### 4.1 Training Curves

The plots below show the training and validation loss and the validation BLEU across epochs.

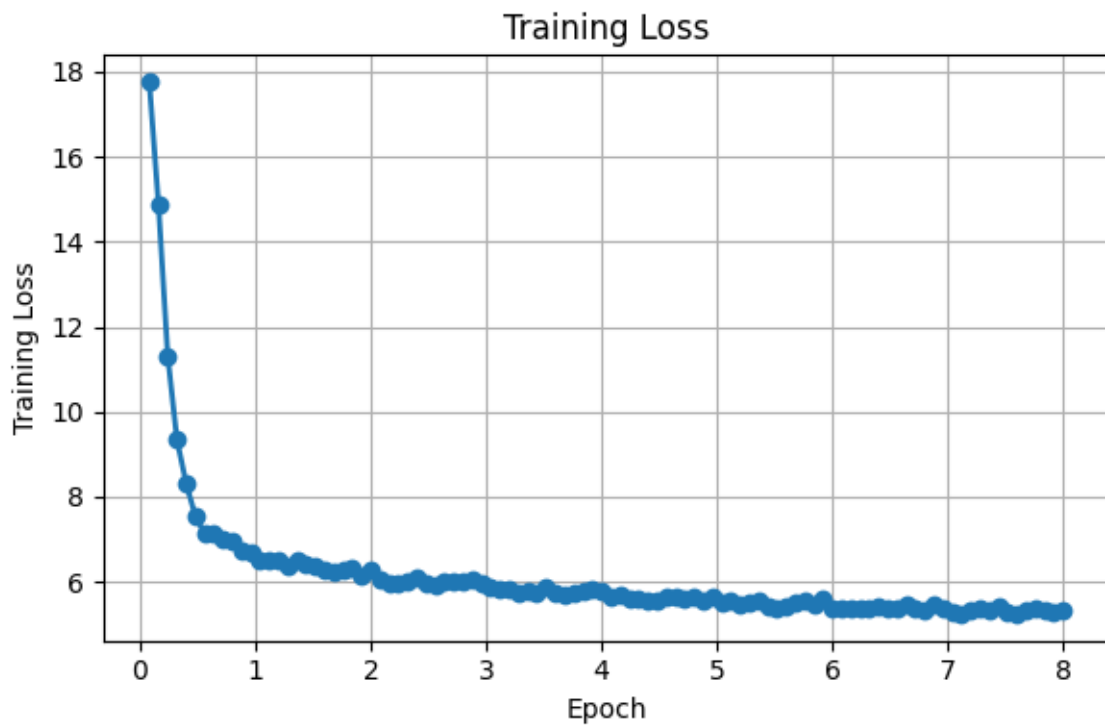


Figure 1. Training Loss over epochs.

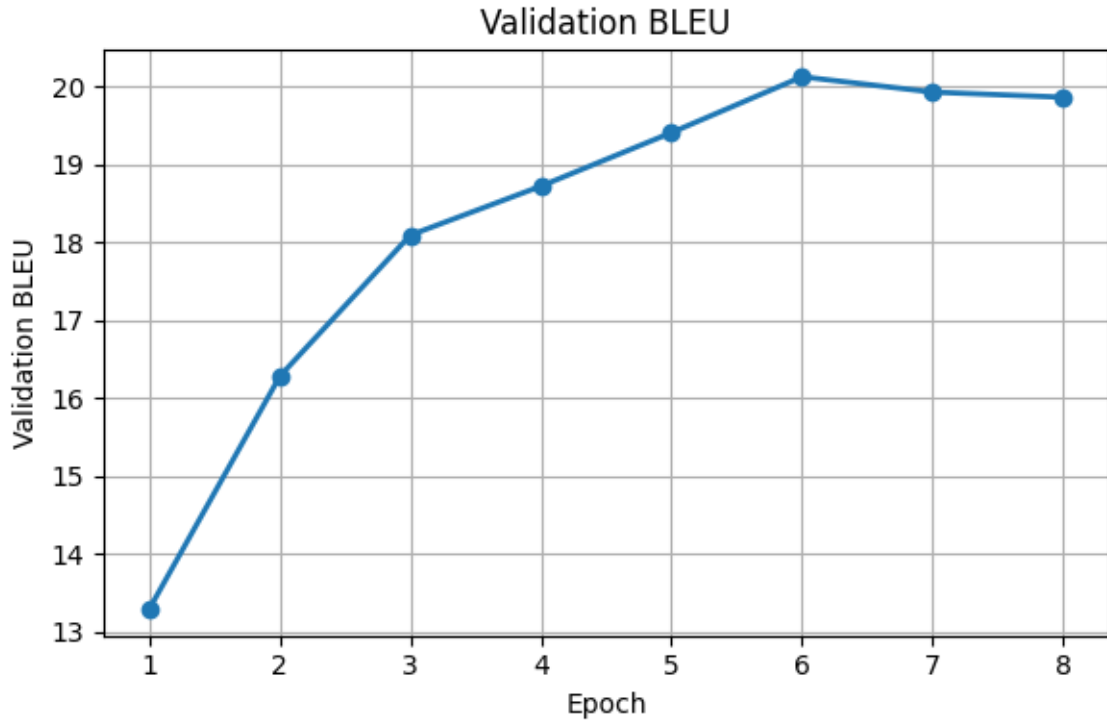


Figure 2. Validation BLEU over epochs.

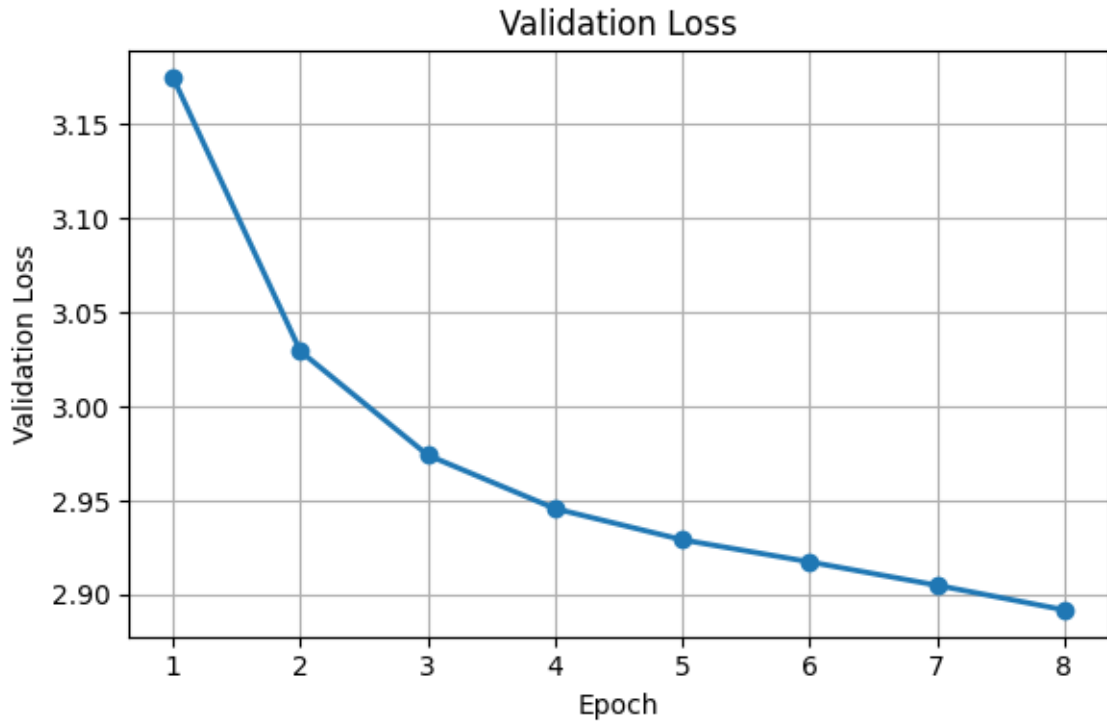


Figure 3. Validation Loss over epochs.

## 5. Translation Examples and Error Analysis

Representative examples from the test-set predictions:

| ID  | Source (Sanskrit)                                  | Reference (English)                                | Prediction (English)                               |
|-----|----------------------------------------------------|----------------------------------------------------|----------------------------------------------------|
| 28  | बालः त्वयि प्रेमं प्रदर्शयति ।                     | Boy displays affection in you.                     | Boy displays affection in you.                     |
| 131 | यह पूरे शरीर में रक्त परिसंचरण में सुधार करता है । | It improves blood circulation throughout the body. | It improves blood circulation throughout the body. |

|     |                                                                                                                        |                                                                                            |                                                                                 |
|-----|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 3   | तदा, तत्स्वयं ड्राइवर निमित्तम् अन्वेष्टयति ।                                                                          | Then it will automatically begin searching for drivers.                                    | Then it will automatically search for the driver.                               |
| 4   | सर्वेभ्यः इटरेशन्-अर्थम्, iterator इतीदं प्रत्येकस्मै इण्डेक्स्-वेल्यू-इत्यस्मै सेट् क्रियते । 1,1 पश्चात् 1,2 एवम्... | The iterator will be set to each of the indices values...                                  | The meaning of iteration for all iterators is set to the index value...         |
| 401 | भिल्लमाचार्य इति अस्यनामान्तरम्।                                                                                       | He was also known as Bhillamacharya.                                                       | His name is Villamacharya.                                                      |
| 867 | डौन्-लोड्-विषये विवरणम्:                                                                                               | Information regarding download:                                                            | Details of the download                                                         |
| 971 | श्वास भरे, बिना किसी दबाव के श्वास भरे । यह श्वास सामान्य रूप से भरा जाना चाहिए ।                                      | Inhale spontaneously and passively. Let air enter the body through the passive inhalation. | Inhale, inhale without any pressure. This inhalation should be filled normally. |

1. **Best cases:** Simple instructional or declarative sentences are translated almost perfectly. For example, IDs 28 and 131 exactly match the reference translations.
2. **Common issue:** The model often preserves the correct meaning while using different wording. For example, ID 3 translates "begin searching for drivers" as "search for the driver", and ID 4 paraphrases the iterator description while retaining the underlying meaning. Such semantic variations reduce BLEU despite producing fluent translations.
3. **Weak spot:** Proper nouns and transliterated names remain challenging. In ID 401, Bhillamacharya is translated as Villamacharya, demonstrating that transliteration errors are more common than grammatical errors.
4. **BLEU limitation:** BLEU penalizes valid paraphrases. For example, ID 867 translates "Information regarding download" as "Details of the download". Although the wording differs, the meaning is essentially the same. Similarly, ID 971 conveys the breathing instruction correctly using a more natural English expression. These cases receive lower BLEU despite being semantically accurate, which is why BERTScore is also reported.

## 6. Experiments

- **Decoding.** I tuned the decoding parameters to improve translation quality. Beam search with a beam size of 6, a length penalty of 1.0, and a maximum decode length of 256 produced more complete and fluent translations than the default decoding settings.
- **Training.** I fine-tuned the pretrained NLLB-200 model using a learning rate of  $2 \times 10^{-5}$ , batch size 8 with gradient accumulation 2, label smoothing 0.1, and weight decay 0.01. The model converged within a few epochs, and early stopping selected the checkpoint with the highest validation BLEU.
- **Evaluation:** Translation quality was evaluated using Corpus BLEU, Sentence BLEU, and BERTScore (F1). BLEU measures n-gram overlap with the reference translations, while BERTScore captures semantic similarity, making it a useful complementary metric for paraphrased translations.

## 7. Discussion

- **Design choices:** I fine-tuned NLLB-200 Distilled (600M) model instead of training from scratch because it already knows Sanskrit and gives much better results on this small dataset. The increase in computation cost is due to the size: at ~615M parameters it is large and slower, which reduces the efficiency score.
- **Challenges:** the data mixes everyday sentences with software and technical terms, and Sanskrit's rich word forms and single references make BLEU look low even when translations are good.
- **Limitations:** only the given data was used (no extra data), the model is large, and BLEU understates real quality - which is why BERTScore is reported too. A smaller model or a method like LoRA could reduce the cost.

## 8. Pre-trained Models Disclosure

- **Translation model:** facebook/nllb-200-distilled-600M, pretrained by Meta and fine-tuned here on the provided data only (a model download, not an API).
- **Tokenizer:** NLLB's own SentencePiece tokenizer, used as-is.
- **BERTScore:** the bert-score library uses a pretrained RoBERTa for evaluation only, it was not used during model training or inference.
- **APIs:** no external translation or inference APIs were used.

## 9. References

1. NLLB Team, No Language Left Behind: Scaling Human-Centered Machine Translation, Meta AI, 2022.
2. Vaswani, A., et al., Attention Is All You Need, Advances in Neural Information Processing Systems (NeurIPS), 2017.
3. Fan, A., et al., Beyond English-Centric Multilingual Machine Translation (M2M-100), Journal of Machine Learning Research (JMLR), 2021.
4. Wolf, T., et al., Transformers: State-of-the-Art Natural Language Processing, EMNLP System Demonstrations, 2020.
5. Papineni, K., Roukos, S., Ward, T., & Zhu, W., BLEU: A Method for Automatic Evaluation of Machine Translation, ACL, 2002.
6. Zhang, T., et al., BERTScore: Evaluating Text Generation with BERT, ICLR, 2020.

## 10. Code and Submission

- **Source code:** The complete implementation is available on GitHub:  
<https://github.com/G25AIT1012/Natural-Language-Understanding-Assignment-2>
- **Repository contents:**
  1. G25AIT1012.ipynb - Complete training and inference notebook.
  2. Report.pdf - Assignment report.
  3. submission.csv - Final predictions on the test set
- **Fine-tuned model:** The trained model (finetuned\_model.zip) is available separately via the following Google Drive link:  
<https://drive.google.com/drive/u/5/folders/1BDJzZ6UUdAqvga8ipGztb95kWjRX2zND>